DUAL-HYPERGRAPH INDEXING: BRIDGING KNOWLEDGE ISLANDS FOR MULTI-HOP REASONING IN RETRIEVAL-AUGMENTED GENERATION

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ABSTRACT

While hypergraph-based Retrieval-Augmented Generation (RAG) effectively captures higher-order multi-entity correlations, existing paradigms treat extracted hyperedges as isolated factual assertions. This structural fragmentation engenders rigid "knowledge islands" that bottleneck multi-hop causal inference, temporal tracking, and narrative synthesis. To systematically address these challenges, we introduce Dual-Hypergraph Indexing (DHI), a hierarchical representation framework that elevates discrete facts into structured analytical insights. DHI couples a foundational entity-relation factual hypergraph (H_K) with an elevated deep-insight hypergraph (H_D) via a dual-pathway aggregation algorithm. Specifically, DHI employs: (1) importance-driven hub aggregation via 5-metric topological profiling and adaptive thresholding to capture spatial semantic clusters; and (2) temporal chunk-chain progressive aggregation via sliding-window greedy exploration to track chronological evolutions. Across five benchmarks, DHI achieves state-of-the-art performance, boosting logical coherence by +1.53 on the multidisciplinary Mix benchmark and scoring 85.78% on complex medical pathology reasoning tasks. DHI provides a robust architecture for next-generation multi-hop RAG.

***Index Terms*—** Retrieval-augmented generation, hypergraph representation, knowledge aggregation

1. INTRODUCTION

The advent of Large Language Models (LLMs) has fundamentally transformed natural language processing, enabling unprecedented capabilities in complex reasoning. However, their parametric memory remains inherently static and susceptible to factual hallucinations, particularly when navigating specialized, proprietary, or rapidly evolving domains [1]. Non-parametric retrieval augmentation—commonly known as Retrieval-Augmented Generation (RAG)—curtails these hallucinations by grounding the generative space in dynamically retrieved, verifiable external evidence [2, 3].

As downstream user queries transition from localized, single-fact retrieval toward multifaceted deductive reasoning and multi-hop synthesis, knowledge representation architectures must evolve. Initial paradigms relying on unstructured text chunk embeddings frequently fail to capture cross-document relational semantics [4]. To address this, structured graph topographies, such as GraphRAG [5] and LightRAG [6], explicitly construct entity-relation knowledge graphs to enable macro-level community summaries. More recently, hypergraph-driven architectures have demonstrated distinct topological advantages. By natively modeling complex, non-pairwise interactions among multiple entities ($n \geq 3$) as unified hyperedges, hypergraph RAG prevents the severe semantic distortion caused by binary edge decomposition [7].

Despite these topological advancements, contemporary hypergraph RAG architectures predominantly exhibit a critical structural limitation we formally term the "Knowledge Island Problem". Specifically, existing pipelines typically extract multi-entity hyperedges independently from localized text chunks [8, 9], inheriting the structural fragmentation common in chunk-based retrieval [5, 10] and treating each hyperedge as a static, isolated semantic silo. Consequently, these architectures lack explicit mathematical pathways to synthesize broader thematic patterns, global causal dependencies, and longitudinal chronological progressions spanning multiple independent hyperedges. During multi-hop retrieval, standard retrievers supply a disconnected subset of assertions [11]. Deprived of pre-computed relational bridges, the generator is forced to infer cross-passage dependencies on the fly—a process highly vulnerable to attention dispersion, cognitive overload, and cascading hallucinations [12, 13].

To transcend these limitations, we propose Dual-Hypergraph Indexing (DHI), a novel hierarchical representation framework driven by a robust dual-pathway knowledge aggregation mechanism. DHI fundamentally decouples the indexing structure into a meticulously coordinated two-tiered hierarchy: a foundational factual hypergraph H_K capturing granular atomic assertions, and an elevated deep insight hypergraph H_D modeling synthesized conceptual schemas.

DHI's core computational engine executes two orthogonal aggregation pathways:

- **Importance-Driven Hub Aggregation (Path A):** Captures spatial topological significance. We evaluate a robust 5-dimensional structural profiling vector. Applying P90-normalization and an adaptive elbow cutoff, DHI identifies and synthesizes systemic cross-context observations centered around core domain hubs.
- **Temporal Chunk-Chain Progressive Aggregation (Path B):** Captures longitudinal narrative evolutions and sequential causality. By tracking source-chunk chronological footprints within a constrained sliding window, DHI perfectly bridges sequential causalities spanning vast document distances.

By seamlessly bridging disconnected knowledge islands, DHI achieves state-of-the-art logical coherence (85.47 on multidisciplinary benchmarks) while significantly reducing context tokens,

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establishing a robust blueprint for reasoning-intensive Graph RAG.

2. RELATION TO PRIOR WORK

2.1. Chunk-based and Graph-based RAG Architectures

Standard RAG systems (e.g., RETRO [14], NaiveRAG) perform dense vector similarity matching over fixed-length text chunks. While efficient, chunking inherently severs relational continuity across paragraph boundaries [10]. To resolve this, GraphRAG [5] pioneered automated entity-relation graphs paired with Leiden community clustering to generate hierarchical summaries. LightRAG [6] further optimized this via a dual-level entity-relation scheme, while HippoRAG [15] introduced neurobiologically inspired memory integration. Moreover, active RAG frameworks [16] have explored multi-turn retrieval paths. Nevertheless, traditional pairwise graphs fundamentally fail to represent high-order interactions involving three or more entities simultaneously, leading to unavoidable structural loss [17].

2.2. Hypergraph Representation in Information Retrieval

Hypergraph Neural Networks [7] mathematically validate that hyperedges preserve non-decomposable group interactions. Subsequent works further introduced hypergraph attention mechanisms to dynamically capture high-order correlation structures. HyperRAG [9] proposed localized hypergraph modeling, while CogRAG [8] introduced cognitive-inspired theme alignment.

Despite these advances, existing hypergraph RAG approaches almost universally treat extracted hyperedges as static structural entities. They fail to distill cross-hyperedge causal trajectories or longitudinal temporal developments [18]. DHI departs from these paradigms by formalizing a dual-hypergraph architecture that explicitly synthesizes higher-order insights via synergistic spatial and chronological exploration, directly neutralizing the "Knowledge Island Problem".

3. PROPOSED METHODOLOGY

3.1. System Architecture and Formal Definition

Given a raw textual corpus partitioned into sequentially ordered, overlapping chunks $\mathcal{C} = \{c_1, \dots, c_M\}$, DHI constructs a hierarchically coupled dual-hypergraph framework, defined as (H_K, H_D):

- **Factual Hypergraph $H_K = (V_K, E_K)$:** The foundational layer. Vertices $v \in V_K$ denote named entities. Hyperedges $e \in E_K$ represent simple pairwise ($|e| = 2$) or complex higher-order ($|e| \geq 3$) relations. Every higher-order hyperedge e encapsulates a specific relation summary and an atomic direct insight $\iota(e)$, generated during the initial extraction phase.
- **Deep Insight Hypergraph $H_D = (V_D, E_D)$:** The synthesis layer. Vertices $u \in V_D$ establish a strict bijective mapping with higher-order hyperedges of H_K, formulated as $V_D = \{\pi(e) \mid e \in E_K, |e| \geq 3\}$. Each deep insight hyperedge $\varepsilon \in E_D$ binds a targeted subset of these insight vertices, enriched with a synthesized narrative insight $\mathcal{I}(\varepsilon)$.

3.2. The Knowledge Island Formulation

The necessity of H_D arises from spatial isolation within H_K. For a multi-hop query q requiring logical traversal from v_a to v_c via intermediary v_b, standard retrieval yields disjoint hyperedges $e_1 = \{v_a, v_b, \dots\}$ and $e_2 = \{v_b, v_c, \dots\}$. The semantic void between $\iota(e_1)$ and $\iota(e_2)$ constitutes the knowledge island boundary. DHI actively computes topological bridges $\varepsilon = \{\pi(e_1), \pi(e_2)\}$ inside H_D prior to retrieval.

Algorithm 1 Dual-Path Dual-Hypergraph Indexing

Require: Sequential Corpus $\mathcal{C}$, Sliding window limit $W = 10$.
Ensure: Factual Hypergraph H_K, Deep Insight Hypergraph H_D.
Parse $\mathcal{C}$ via LLM to extract V_K, E_K, and direct atomic insights $\iota(e)$.
Initialize $H_K \leftarrow (V_K, E_K)$.
Initialize $H_D \leftarrow (V_D = \{\pi(e) \mid e \in E_K, |e| \geq 3\}, E_D = \emptyset)$.
/* Path A: Importance-Driven Hub Aggregation */
for each unique entity $v \in V_K$ **do**
    Compute $\mathbf{m}(v)$ via Eq. (1) and robust $S(v)$ via Eqs. (2)–(3).
end for
$V_{\text{hub}} \leftarrow \text{ElbowCutoff}(\text{SortDescending}(V_K, S))$.
for each identified hub entity $v^* \in V_{\text{hub}}$ **do**
    $\mathcal{I}_{\text{hub}} \leftarrow \text{LLM_Synthesize_Hub}(v^*, E(v^*))$.
    $E_D \leftarrow E_D \cup \{(\{\pi(e) \mid e \in E(v^*)\}, \mathcal{I}_{\text{hub}})\}$.
end for
/* Path B: Temporal Chunk-Chain Aggregation */
Chains $\leftarrow \text{GreedyBranchingSearch}(E_K, W)$ using Eq. (5).
$\text{Chains}_{\text{pruned}} \leftarrow \text{FilterAndDeduplicate}(\text{Chains}, L_{\min} = 3)$.
for each valid temporal chain $\mathcal{X} \in \text{Chains}_{\text{pruned}}$ **do**
    $\mathcal{I}_{\text{chain}} \leftarrow \text{LLM_Synthesize_Temporal}(\mathcal{X})$.
    $E_D \leftarrow E_D \cup \{(\{\pi(e) \mid e \in \mathcal{X}\}, \mathcal{I}_{\text{chain}})\}$.
end for
Construct dense vector database indices for H_K and H_D.
return H_K, H_D

3.3. Dual-Path Insight Aggregation Algorithm

DHI executes two orthogonal aggregation pathways (Algorithm 1) targeting distinct axes of information distribution.

3.3.1. Path A: Importance-Driven Hub Entity Aggregation

Path A identifies authoritative semantic hubs. Because vertex degree alone is structurally insufficient for evaluating prominence within hypergraphs [19], we formulate a 5-dimensional structural profiling vector for candidate entities $v \in V_K$:

$$\mathbf{m}(v) = \left[d(v), |E(v)|, \bar{s}(v), w(v), |N(v)|\right]^T \tag{1}$$

Here, $d(v)$ is vertex degree, $|E(v)|$ signifies incident hyperedge count, $\bar{s}(v) = |E(v)|^{-1} \sum_{e \in E(v)} |e|$ represents average hyperedge scale, $w(v)$ is cumulative semantic weight, and $|N(v)|$ tracks distinct neighbor coverage.

To prevent extreme high-degree generic outliers from statistically obfuscating the true semantic backbone, we implement a P90-robust normalization strategy:

$$\tilde{m}_k(v) = \frac{\min(m_k(v), P_{90}(m_k)) - m_k^{\min}}{P_{90}(m_k) - m_k^{\min}} \tag{2}$$

where $k \in \{1, \dots, 5\}$, $m_k^{\min} = \min_u m_k(u)$, and P_{90} is the 90th percentile. The composite importance score is the uniform mean:

$$S(v) = \frac{1}{5} \sum_{k=1}^{5} \tilde{m}_k(v) \tag{3}$$

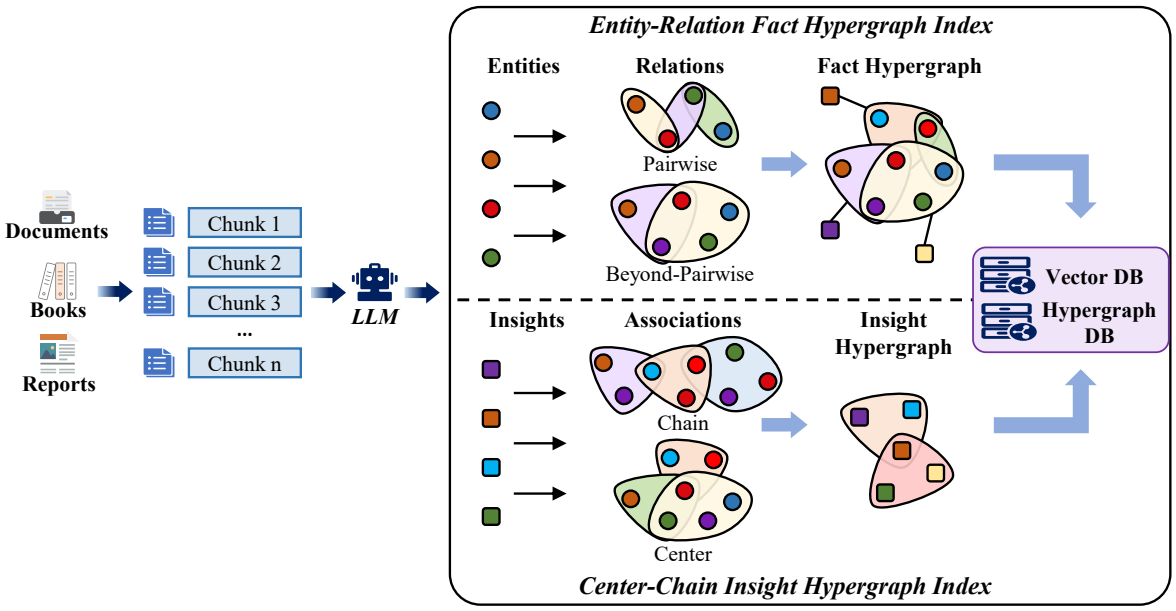

Fig. 1. The end-to-end operational workflow of the Dual-Hypergraph Indexing (DHI) framework. Source documents are iteratively segmented into sequential sliding chunks. An LLM extracts multi-entity hyperedges equipped with atomic direct insights to construct the foundational factual hypergraph $\mathcal{H}_K$ (bottom). Dual pathways—topological hub aggregation (Path A) and temporal chunk-chain progressive aggregation (Path B)—synthesize these assertions into high-order semantic correlations, populating the deep insight hypergraph $\mathcal{H}_D$ (top). Bidirectional mapping ensures rigorous cross-layer provenance, mitigating hallucinations during context assembly.

Entities are sorted in descending order of $S(v)$. We autonomously determine the optimal hub inclusion boundary index k^* via maximum first-order difference, bounded by a 20% system safety cap to prevent insight dilution:

$$k^* = \arg\max_k \big(S(v_k) - S(v_{k+1})\big), \quad \text{s.t. } k \le 0.20 \cdot |V_K| \tag{4}$$

For each identified hub v^*, an LLM synthesizes its entire incident hyperedge set $E(v^*)$ into a systemic hub insight $\mathcal{I}_{\text{hub}}(v^*)$. The insight hyperedge $\varepsilon_{\text{hub}} = \{\pi(e) \mid e \in E(v^*)\}$ is instantiated within H_D.

3.3.2. Path B: Temporal Chunk-Chain Progressive Aggregation

While Path A extracts static radial clusters, Path B discovers longitudinal narrative chains and chronologies. Let $\tau(c) \in \mathbb{N}^+$ denote the absolute sequential position of chunk c. For any hyperedge e, its temporal footprint is $\tau(e) = \{\tau(c) \mid c \in \text{source}(e)\}$.

We initialize potential analytical chains and iteratively expand them via a greedy branching algorithm within a sliding window W. A candidate hyperedge e' is appended to an active evolving chain $\mathcal{X} = (e_1, \ldots, e_L)$ if three strict conditions are met:

$$\begin{cases} \min \tau(e') - \max \tau(e_L) \le W, & \text{(Temporal Proximity)} \\ |e_L \cap e'| \ge 1, & \text{(Entity Overlap)} \\ \max\limits_{v \in \mathcal{V}(\mathcal{X})} \text{Freq}(v) < \dfrac{2}{3}, & \text{(Anti-Hub Penalty)} \end{cases} \tag{5}$$

where $\mathcal{V}(\mathcal{X})$ denotes the collective entity union of the chain, and $\text{Freq}(v)$ is the fraction of hyperedges containing v.

Crucially, the $2/3$ dominance threshold (Anti-Hub Penalty) explicitly guarantees that Path B remains mathematically orthogonal to Path A. It actively penalizes sequences that merely orbit a single central entity, forcing the algorithm to trace true linear narrative evolutions rather than topologically degenerating into localized hub clusters. Chains satisfying minimum length $L \ge 3$ are retained, and redundant sub-chains are aggressively pruned. An LLM synthesizes each valid chain into $\mathcal{I}_{\text{chain}}$, forming a temporal hyperedge $\varepsilon_{\text{chain}}$ in H_D.

3.4. Cross-Layer Joint Retrieval and Context Assembly

Upon receiving a complex user query q, DHI executes a highly coordinated, multi-scale semantic retrieval cascade:

1. **Factual Subgraph Retrieval:** Compute dense query-entity similarity $s(v, q) = \frac{\mathbf{e}_v \cdot \mathbf{q}}{\|\mathbf{e}_v\| \|\mathbf{q}\|}$. Top-$k$ scoring entities induce an active factual hyperedge subset E_{sub}.
2. **Topological Projection and Diffusion:** Utilizing the deterministic mapping π, active hyperedges project to seed insight vertices $U_{\text{sub}} = \{\pi(e) \mid e \in E_{\text{sub}}, |e| \ge 3\}$. A rapid 1-hop traversal on H_D harvests deep insights $\mathcal{E}^*$.
3. **Structured Context Formatting:** Granular facts from E_{sub} and overarching narrative summaries from $\mathcal{E}^*$ are concatenated into a unified prompt context $\mathcal{P}(q)$ for final generation.

4. EXPERIMENTAL EVALUATION

4.1. Experimental Setup and Implementation Details

Datasets: We rigorously validate DHI across five multi-domain public benchmarks testing basic retrieval to multi-step causal etiology: Mix (multidisciplinary, 61 documents, 560 chunks), CS (Computer

Table 1. Overall composite performance comparison across five multi-domain benchmarks (0–100 scale, higher is better). Agri., Neuro., and Patho. denote Agriculture, Neurology, and Pathology, respectively.

Method	Mix	CS	Agri.	Neuro.	Patho.
LLM (Zero-shot)	79.30	81.08	79.64	81.15	82.81
NaiveRAG [2]	78.09	79.43	76.20	79.20	82.04
GraphRAG [5]	81.06	84.03	79.98	83.10	82.72
LightRAG [6]	81.01	81.25	79.05	81.82	84.43
HiRAG [21]	82.85	83.33	81.19	82.71	84.13
Hyper-RAG [9]	80.39	83.88	81.98	83.74	84.41
DHI (Ours)	**83.18**	**84.66**	**82.97**	**84.23**	**85.78**

Table 2. Granular capability breakdown evaluating five specific NLP dimensions on the Mix benchmark.

Method	Comp.	Diver.	Empo.	Logi.	Read.
LLM	85.40	73.80	73.76	81.54	82.00
NaiveRAG [2]	85.00	72.36	71.64	81.50	79.96
GraphRAG [5]	87.70	77.10	75.10	83.58	81.82
LightRAG [6]	88.00	78.10	74.82	82.38	81.76
HiRAG [21]	**90.00**	80.00	**77.66**	83.94	82.64
Hyper-RAG [9]	84.10	78.48	74.68	83.16	81.54
DHI (Ours)	88.98	**80.20**	77.35	**85.47**	**83.92**

Science, 10 docs, 1992 chunks), Agriculture (12 docs, 1813 chunks), Neurology (medical textbook, 1790 chunks), and Pathology (clinical causality, 824 chunks) [20].

Baselines: We benchmark DHI against six strong baselines: LLM (zero-shot), NaiveRAG [2], GraphRAG [5], LightRAG [6], HiRAG [21], and Hyper-RAG [9]. Systems universally deploy GPT-4o-mini for generation and `text-embedding-3-small` for embeddings ($T = 0$). Chunking size is 500 tokens with 50-token overlap. DHI sets $W = 10$. Evaluations follow LLM-as-a-Judge protocols across five dimensions (0–100 scale).

4.2. Main Results and Domain Adaptability

As detailed in Table 1, DHI achieves state-of-the-art composite metrics across all domains. On the Mix benchmark, DHI records 83.18, outperforming Hyper-RAG (80.39) by +2.79 absolute points and surpassing the advanced HiRAG architecture (82.85).

Performance gains are most acutely notable in domains requiring intensive deductive reasoning. On the Pathology dataset—where medical causality spans multiple chapters—DHI achieves a leading score of 85.78, outstripping LightRAG (84.43) and GraphRAG (82.72) by 1.35 and 3.06 points. This firmly verifies that pre-synthesizing higher-order topological insights provides highly reliable inductive support [22].

4.3. Dimensional Analysis and Logical Coherence

Table 2 provides a granular capability breakdown on the Mix benchmark. DHI achieves undeniably superior results in Logicality (85.47), Readability (83.92), and Diversity (80.20).

The substantial +1.53 margin in Logicality over HiRAG demonstrates that mathematically presenting structured causal progressions relieves the generator LLM from making unassisted inferential leaps over disjointed contexts [23]. While HiRAG achieves a higher Comprehensiveness score, it accomplishes this by exhaustively concatenating macro-level community summaries, inadvertently introducing massive non-essential context. Conversely, DHI prioritizes strict structural relevance, deliberately avoiding token bloat.

4.4. Ablation Study: The Vital Synergy of Dual Pathways

We evaluate ablative variants of the DHI architecture directly on Mix:

- **w/o Path A (No Hubs):** Removing topological hub aggregation causes the composite score to precipitously decline to 81.74, and Logicality drops to 82.80. Without spatial distillation, central themes remain fragmented.
- **w/o Path B (No Temporal Chaining):** Omitting sequence chaining yields a suppressed composite of 82.25. This proves that explicit sequential chaining is absolutely essential for tracing longitudinal narrative evolutions across distant chunks.
- **Full DHI Architecture:** Unifying both pathways effortlessly attains the maximal 83.18, verifying that hub clustering and sequential progression are entirely orthogonal yet mutually reinforcing inductive biases.

4.5. Qualitative Case Study

Consider a multi-hop query in Pathology: *"How does prolonged exposure to Agent X subsequently trigger the cascade leading to Syndrome Z?"* Hyper-RAG retrieves isolated hyperedges (Agent X causes degradation; degraded pathways cause Syndrome Z) from distinct chapters. Under DHI, Path B proactively instantiates an overarching hyperedge $\varepsilon_{\text{chain}} \in H_D$ during indexing that synthesizes this exact etiology. During retrieval, the model instantly pulls this verified chronological chain, providing a flawless, hallucination-free logical bridge.

5. DISCUSSION AND LIMITATIONS

While DHI achieves robust reasoning coherence, it inherently relies on the absolute precision of the LLM during the initial entity extraction phase. Factual misinterpretations risk being algorithmically amplified into H_D. Future iterations must investigate dynamic confidence-weighting algorithms and self-corrective feedback mechanisms [24] to mathematically guarantee the strict fidelity of synthesized schemas.

6. CONCLUSION

We comprehensively expose and resolve the "Knowledge Island Problem" bottlenecking existing hypergraph-based RAG architectures. By innovatively decoupling knowledge representations into a foundational factual hypergraph H_K and an elevated deep insight hypergraph H_D, Dual-Hypergraph Indexing (DHI) natively captures both static topological prominence and highly dynamic chronological evolutions. Through synergistic dual-pathway aggregation, DHI synthesizes isolated facts into cohesive analytical insights, elevating logical coherence by +1.53 points over current systems. Moving forward, future research will actively investigate recursive insight hierarchies and lightweight edge deployments to democratize robust multi-hop reasoning.

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